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Cesium-131 Brachytherapy in Patients that are Not Candidates for Post-Operative External Beam Radiation Therapy: Interim Results of a Single Arm Prospective Study

B. Parashar,¹ L. Pereira,² D. Kamdar,² G. Har-El,³ L. Krevitt,⁴ B.A. Miles,⁵ L. Wang,⁶ F. Tran,⁷ D. Knobel,⁸ D. Hirsch,⁷ and D. Frank,⁸; ¹Northwell, Lake Success, NY, ²Northwell Health, New Hyde Park, NY, ³Northwell Health Lenox Hill Hospital, New York, NY, ⁴Lenox Hill Hospital, Northwell Health, New York, NY, ⁵Department of Otolaryngology, Northwell Health, New York City, NY, ⁶Department of Radiation Medicine, Northwell, Lake Success, NY, ⁷Northwell Health, New York, NY, ⁸Northwell Health, Lake Success, NY

Purpose/Objective(s): Patients with advanced head and neck cancers (HNC) that present with resectable disease but are not candidates for post-operative external beam radiation pose a unique challenge. Reasons for ineligibility may include poor-performance status, prior-radiation therapy, or patient choice. We report an interim report of toxicity and outcomes of our institutional prospective study using Cesium-131 (Cs-131) brachytherapy at the time of surgery.

Materials/Methods: This is an IRB approved prospective single arm study. Eligibility includes operable advanced HNC needing post-operative external beam radiation (EBRT). Reasons for deciding against EBRT post-operatively included prior full course RT, patient choice, and performance status. Surgery includes gross total resection followed by Cs-131 implantation of the tumor bed, seeds 1 cm apart. Immediate reconstruction was performed if necessary. All patients were evaluated every 2-4 months for the first 12 months, followed by evaluation every 3-6 months. Recurrence was evaluated by clinical examination, contrast CT and/or FDG-PET. Toxicity was assessed using CTCAE v.5 criteria. Survival estimates were performed using the Kaplan-Meier method. Local control was defined as no recurrence within 0.5cm of the implant site.

Results: Eleven patients have been accrued till date in this study. Median age is 73 (Range 60-95). There were 3 females (27%) and 8 males (73%). Primary tumor sites are 7 (63%) oral cavity, 2 (18%) oropharynx, 1 larynx and 1 salivary gland. Nine patients had Cs-131 neck implant while 2 patients primary site implanted. Median prescription dose was 80Gy (Range:70-80Gy). Six patients had prior RT to the implanted site. Seven (63%) patients were treated for recurrent disease. With a median follow-up of 8 months (Range 1-19), 11/11 (100%) patients had local control at the implanted site, 4/7 (37%) patients developed regional recurrence, 2 patients developed distant recurrence. At the last follow-up 8/11 (72%) are alive. There were four patients with grade 2 toxicity including abscess formation, flap dehiscence, skin breakdown, fistula (not ascribed to seeds) all of which were managed conservatively, and fistula was repaired. One patient had a short follow up of 3 months and was reported to have expired although the cause could not be ascertained.

Conclusion: This is an interim report of a prospective study using Cs-131 in HNC that were not candidates for curative post-operative EBRT. The high local control rates and limited toxicity are encouraging. We continue to accrue patients in this study and will report longer term outcomes of this prospective study soon.

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Management of Oral Cavity Squamous Cell Carcinoma in Elderly Patients Using Diffusing Alpha-Emitter Radiation Therapy

A. Popovtzer,¹ Y. Fleissig,² N. Kurman,³ A. Mizrahi,⁴ E. Rosenfeld,³ G. Feliciani,⁵ S.R. Bellia,⁶ D. Silvern,³ A. Sarnelli,⁵ J. Feldman,¹ J. Elia,² N. Hirshoren,² and R. Den,⁷; ¹Hadassah Medical Center, Jerusalem 91120, Israel, ²Hadassah Medical Center, Jerusalem, Israel, ³Rabin Medical Center,

Petach Tikva, Israel, ⁴Department of Head and Neck Surgery, Rabin Medical Center, Beilinson Campus, Petach Tikva, Israel, ⁵Istituto Scientifico Romagnolo per lo Studio E la Cura Dei Tumori - IRST, Medical Physics Department, Meldola (FC), Italy, ⁶IRCCS - Istituto Scientifico Romagnolo per lo Studio E la Cura dei Tumori - IRST, Radiotherapy Department, Meldola (FC), Italy, ⁷Alpha Tau Medical, Jerusalem, Israel

Purpose/Objective(s): The standard treatment for oral cavity squamous cell carcinoma (OSCC) has remained surgery for decades and the accepted approaches are often challenging in older patients. The present research evaluated the efficacy and safety of Diffusing Alpha-Emitter Radiation Therapy (DaRT) as a novel method of treating these solid tumors using alpha-particle interstitial radiotherapy.

Materials/Methods: Data from patients with OSCC were collected from three prospective clinical trials at multiple medical centers in Israel and Italy. Tumor response was determined by Response Evaluation Criteria in Solid Tumors guidelines (RECIST v1.1). Safety was assessed according to Common Terminology Criteria for Adverse Events (CTCAE) V5.0.

Results: Twenty-three patients (11 female and 12 male) were treated with DaRT for OSCC across the participating clinical sites. Subjects were elderly, with a mean age of 85 (range 60-102), with 78% having a cardiovascular history and 43% with diabetes. Tumor locations included the lip, tongue and buccal mucosa, with mean baseline tumor volume of 3.1 cubic cm. All procedures were performed under local anesthesia in an outpatient setting. The objective response rate was 100% for the 19 tumors which were evaluated (the remaining cases were withdrawn prior to evaluation due to unrelated serious adverse events, withdrawal of consent or election to proceed with alternative therapy), with 18 (95%) showing a complete response and 1 (5%) a partial response. All related adverse events were Grade 1 or 2.

Conclusion: All patients tolerated DaRT therapy with 95% complete response rate and favorable safety profile. DaRT has the potential to serve as a treatment for OSCC, particularly in the elderly population.

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Gantry-Based Pencil Beam Scanning (PBS) Proton Therapy for Uveal Melanoma: IMPT vs. Proton Arc Therapy

H. Qi,¹ L. Hu,² S. Huang,³ Y.P. Lee,⁴ F. Yu,¹ Q. Chen,¹ Y. Yang,⁵ M. Kang,¹ H. Zhai,¹ A. Shim,¹ P. Park,¹ X. Ding,⁶ J. Zhou,⁷ C.B. Simone, II¹ C.A. Barker,⁵ and H. Lin¹; ¹New York Proton Center, New York, NY, ²Inova Schar Cancer Institute, Fairfax, VA, ³Tianjin Medical University Cancer Institute & Hospital, Tianjin, Tianjin, China, ⁴University of Iowa Health Care, Iowa City, IA, ⁵Memorial Sloan Kettering Cancer Center, New York, NY, ⁶Department of Radiation Oncology, Corewell Health William Beaumont University Hospital, Royal Oak, MI, ⁷Department of Radiation Oncology, Emory University, Atlanta, GA

Purpose/Objective(s): Ocular proton therapy is often delivered in a dedicated scattering beamline. This study reports multi-field PBS intensity-modulated proton therapy (IMPT) for uveal melanoma (UM) in a regular clinical PBS gantry room. We hypothesize that volumetric modulated proton arc therapy (pArc) can offer superior conformity in comparison to IMPT